

**Jumping on demand**

Scientists have managed to train a spider to jump on demand to discover the secrets of animal movement. <https://digit.in/jun18-B40-1>

**Robot can dress people**

Georgia Tech engineers have built a robot that can slide hospital gowns on people through a sense of touch. <https://digit.in/jun18-B40-2>

Developing the neural lace



It is the year 2028. Anupam, a researcher in a lab working on a neural lace, to extract and implant thoughts and memories, provides an outline of the work done by the company so far and the future

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NEURAL IMPLANTS

What we have achieved so far is a range of input and output prostheses that treat various kinds of neural diseases and conditions. The earliest neural prosthetic devices were implants that fixed a problem with a sensory organ. We developed implants for the deaf or the blind. These tiny devices implanted within the retina or the ear, could simulate the electrical signals that the real organs would send, and the brain got to see or hear through these devices. Initially, these devices were crude, and only approximated what a person fully capable of sight and hearing could see and hear. Over the course of the years, the resolution of these devices have been significantly improved. The sights and sounds provided to the human brain through these devices are far better than they were a few years ago, but they are still nowhere close to the sheer amount of information supplied to the brain by real sense organs. I highly suspect that these bionic implants will ever

have to be as good as the real thing, but every improvement has allowed the deaf and blind to hear more and see more. As Richard Dawkins pointed out in 2006, it is better to have half an eye than no eye at all.

The other major difference we have had is the use of neural implants for the paralysed. There are a number of different causes for paralysis, and we have developed implants that similarly help them in different ways. A paralysed person can choose to interact with computers and smart devices through our implants. They can control wheelchairs, and manipulate robotic arms, by merely thinking about it. The same set of implants have also been used by amputees, to control prosthetic arms and legs. Thanks to the development of soft robotics and sensors, the newer prosthetic arms even have a sense of touch in them. In a set of clinical trials conducted last year, we managed to successfully communicate with a man who has been in coma for the past 27 years. Another set of

implants we have created treat a different range of conditions. This involves electrodes in the heart that connect to the brain, and trigger in certain conditions. Because of these implants, those with Parkinson's disease do not experience tremors, and epileptic fits are neutralised. The same devices are used to treat people with PTSD or OCD.

There are also some experimental devices in the wild, built by the bio-hacking community, using 3D printers and open source schematics, to create some weird and wonderful neural prosthetics. There is a device that allows you to perceive in the ultraviolet and infrared ranges as well, but it reduces the number of colours you can see in the visible spectrum. Most objects designed for humans appear eerie and alien when these devices are used, but it is useful to see through obstacles such as smoke and smog. Some interfaces meant for video gaming enhance your twitch reactions, and reduce the lag in communicating your intentions through input devices such